

Installing Minikube

MacOS

```
brew install minikube  
brew unlink minikube  
brew link minikube
```

```
curl -LO https://github.com/kubernetes/minikube/releases/latest/download/  
minikube-darwin-amd64
```

```
sudo install minikube-darwin-amd64 /usr/local/bin/minikube
```

```
curl -LO https://github.com/kubernetes/minikube/releases/latest/download/  
minikube-darwin-arm64
```

```
sudo install minikube-darwin-arm64 /usr/local/bin/minikube
```

Linux

```
curl -LO https://github.com/kubernetes/minikube/releases/latest/download/  
minikube-linux-amd64
```

```
sudo install minikube-linux-amd64 /usr/local/bin/minikube && rm minikube-linux-  
amd64
```

```
curl -LO https://storage.googleapis.com/minikube/releases/latest/  
minikube_latest_amd64.deb
```

```
sudo dpkg -i minikube_latest_amd64.deb
```

```
curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-  
latest.x86_64.rpm
```

```
sudo rpm -Uvh minikube-latest.x86_64.rpm
```

Windows

```
New-Item -Path 'c:\' -Name 'minikube' -ItemType Directory -Force
```

```
$ProgressPreference = 'SilentlyContinue'; Invoke-WebRequest -OutFile 'c:\minikube\minikube.exe' -Uri 'https://github.com/kubernetes/minikube/releases/latest/download/minikube-windows-amd64.exe' -UseBasicParsing
```

```
$oldPath = [Environment]::GetEnvironmentVariable('Path',  
[EnvironmentVariableTarget]::Machine)
```

```
if ($oldPath.Split(';') -notcontains 'C:\minikube'){  
[Environment]::SetEnvironmentVariable('Path', $('{0};C:\minikube' -f $oldPath),  
[EnvironmentVariableTarget]::Machine)  
}
```

```
winget install Kubernetes.minikube
```

```
choco install minikube
```

Starting and managing Kubernetes clusters

```
minikube start
```

```
minikube status
```

```
kubectl get nodes
```

```
minikube dashboard
```

```
minikube pause
```

```
minikube unpause
```

```
minikube stop
```

```
minikube config set memory 8192
```

```
minikube config set cpus 4
```

```
minikube config set disk-size 100
```

```
minikube delete
```

```
minikube delete --all
```

```
minikube addons list
```

minikube addons enable ingress
minikube addons disable ingress

Basic Cluster Operations

kubectl create deployment nginx --image=nginx

kubectl get nodes

kubectl get pods

kubectl get deployments

kubectl get services

kubectl expose deployment nginx --type=NodePort --port=80

minikube service nginx

kubectl scale deployment nginx --replicas=3

kubectl set image deployment/nginx nginx=nginx:latest

kubectl delete deployment nginx

kubectl delete service nginx

kubectl logs <pod name>

Common issues and troubleshooting

Minikube cluster fails to start

minikube start --driver=docker

minikube config set memory 8192

minikube config set cpus 4

minikube start --alsologtostderr -v=7

Kubectl cannot connect to cluster

minikube status

minikube stop

minikube start

kubectl config use-context minikube

Pods in pending or CrashLoopBackOff

kubectl describe pod <pod-name>

kubectl logs <pod-name>

Service not available

kubectl get service

minikube service <service name>

Disk space or memory errors

minikube delete -all

minikube dashboard

Kind

Installing Kind

MacOS

brew install kind

```
curl -Lo ./kind https://kind.sigs.k8s.io/dl/latest/kind-$(uname)-amd64
```

```
chmod +x ./kind
```

```
sudo mv ./kind /usr/local/bin/kind
```

Linux

```
curl -Lo ./kind https://kind.sigs.k8s.io/dl/latest/kind-$(uname)-amd64
```

```
chmod +x ./kind
```

```
sudo mv ./kind /usr/local/bin/kind
```

Windows

```
curl.exe -Lo kind.exe https://kind.sigs.k8s.io/dl/latest/kind-windows-amd64
```

```
scoop install kind
```

Creating and managing clusters

```
kind create cluster --name demo-cluster
```

```
kind get clusters
```

```
kubectl config use-context kind-kind
```

```
kubectl config get-contexts
```

```
kind delete cluster --name demo-cluster
```

Multi-node cluster setup

```
#kind-multinode-config.yaml
kind: Cluster
apiVersion: kind.x-k8s.io/v1alpha4
nodes:
- role: control-plane
- role: worker
- role: worker
- role: worker
```

```
kind create cluster --name kind-multi-node --config kind-multinode-config.yaml
```

Common issues and troubleshooting

Kubectl cannot connect to cluster

```
kind get kubeconfig --name kind
kind export kubeconfig
```

```
kubectrl config use-context kind-kind
```

Ingress not working

```
kubectrl apply -f https://raw.githubusercontent.com/kubernetes/ingress-nginx/  
controller-v1.9.4/deploy/static/provider/kind/deploy.yaml
```

Image pull errors

```
kind load docker-image my-image:tag
```

K3d

Installing K3d

MacOS

```
brew install k3d
```

```
curl -s https://raw.githubusercontent.com/k3d-io/k3d/main/install.sh | bash
```

Linux

```
curl -s https://raw.githubusercontent.com/k3d-io/k3d/main/install.sh | bash
```

Windows

```
scoop install k3d
```

```
k3d version
```

Creating and managing clusters

```
k3d cluster create cluster1
```

```
kubectrl get nodes
```

```
k3d cluster list
```

```
kubectl config use-context k3d-cluster1
```

```
kubectl config get-contexts
```

```
k3d cluster delete cluster1
```

```
k3d cluster create multi-server --servers 3
```

```
k3d node create server1 --cluster multi-server --role server
```

Common issues and troubleshooting

Kubectl cannot connect to the cluster

```
k3d kubeconfig merge cluster1
```

```
kubectl config current-context
```

```
kubectl config use-context k3d-cluster1
```

Image pull errors

```
k3d image import my-image:tag -c cluster1
```

DNS resolution failure

```
k3d cluster stop cluster1
```

```
k3d cluster start cluster1
```

